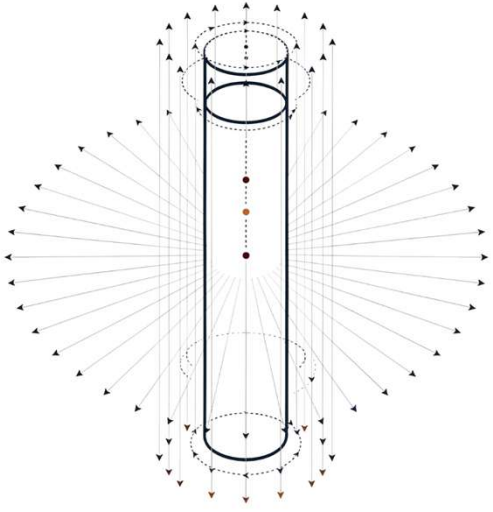
Electric Field and Potential — Fundamentals



Electric Field of a Long Charged Conductor

A conductor with linear charge density q (C/m) produces a **radial electric field** at distance r:

> **$E = q / (2\pi\varepsilon r)$** (V/m)

where $\varepsilon = \varepsilon_0 = 8.854\times10^{-12}$ F/m for air. The field is directed radially outward for positive charge.

Potential Difference Between Two Points

Integrating the electric field from r_1 to r_2:

> **$V12 = (q / 2\pi\varepsilon) \cdot \ln(r_2 / r_1)$** (V)

Superposition for Three-Phase Lines

The voltage at any conductor is the algebraic sum of contributions from all three phase charges qa, qb, qc. For a balanced system: qa + qb + qc = 0, which simplifies the expression significantly.